

OLANDS NORRA UDDE, Sweden

WMO: 025750

Lat: 57.3671N

Lon: 17.0954E

Elev: 10

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	20.5	23.3	14.1	11.3	23.0	16.9	12.9	25.3	36.5	28.9	33.4	32.0	12.5	340	0.689

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	77.1	67.0	74.3	66.0	72.0	64.7	69.3	74.2	67.7	72.2	66.1	70.6	10.7	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.5	101.1	72.1	65.8	95.2	70.5	64.2	89.9	69.1	33.4	74.3	32.0	72.3	30.8	70.8	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 32.9	28.6	25.1	DB	17.8	82.6	4.7	2.7	14.4	84.5	11.6	86.1	9.0	87.6	5.5	89.5	(4)
(5)			WB	16.7	71.1	4.5	2.6	13.5	73.0	10.9	74.6	8.3	76.0	5.1	77.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	47.6	33.4	33.2	36.2	42.1	50.5	59.6	65.4	65.0	58.1	48.9	41.7	36.5	(6)
(7)		DBStd	12.56	5.01	5.40	5.43	4.91	5.39	4.75	4.34	4.05	4.21	4.71	4.47	5.33	(7)
(8)		HDD50	2461	516	470	428	240	59	1	0	0	1	77	249	420	(8)
(9)		HDD65	6458	981	890	892	686	448	172	48	50	209	499	698	885	(9)
(10)		CDD50	1599	0	0	0	5	76	288	477	465	244	43	1	0	(10)
(11)		CDD65	120	0	0	0	0	0	9	60	49	2	0	0	0	(11)
(12)		CDH74	275	0	0	0	0	2	36	161	76	0	0	0	0	(12)
(13)		CDH80	22	0	0	0	0	0	4	13	5	0	0	0	0	(13)
(14)	Wind	WSAvg	12.7	14.9	14.1	13.1	11.8	10.8	10.9	10.5	10.8	12.3	13.8	14.4	15.0	(14)
(15)	Precipitation	PrecAvg	16.6	1.1	0.8	0.9	1.0	1.2	1.4	1.7	1.9	2.0	1.4	1.8	1.5	(15)
(16)		PrecMax	26.1	2.6	2.1	2.5	2.8	3.1	3.1	4.0	4.8	5.2	4.7	4.2	3.3	(16)
(17)		PrecMin	12.1	0.0	0.2	0.1	0.2	0.1	0.0	0.2	0.3	0.6	0.1	0.4	0.2	(17)
(18)		PrecStd	2.8	0.7	0.4	0.5	0.6	0.7	0.8	1.0	1.2	1.2	0.9	0.9	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.4	46.9	54.7	59.8	70.3	78.7	81.9	79.7	72.1	61.6	53.8	50.1	(19)
(20)			MCWB	43.7	43.3	45.9	51.1	57.6	66.0	70.3	69.2	64.5	56.8	51.2	47.8	(20)
(21)		2%	DB	43.8	44.1	49.2	55.6	64.6	73.4	78.1	75.9	69.1	58.6	51.1	47.0	(21)
(22)			MCWB	41.7	41.2	43.3	48.6	55.6	62.8	67.7	67.2	62.5	55.5	48.8	44.8	(22)
(23)		5%	DB	41.6	42.2	45.6	52.5	61.3	70.0	75.2	73.5	66.5	56.8	49.3	45.0	(23)
(24)			MCWB	39.6	39.8	41.4	46.7	54.0	61.3	66.6	66.2	60.9	54.2	47.2	43.1	(24)
(25)		10%	DB	40.0	40.3	43.1	49.4	58.7	67.2	72.6	71.4	64.2	55.2	47.7	43.2	(25)
(26)			MCWB	38.3	38.0	39.8	44.8	52.6	59.8	65.5	65.1	59.6	52.8	46.0	41.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.4	44.0	47.2	52.5	59.8	66.9	72.6	71.8	66.6	58.2	51.8	47.9	(27)
(28)			MCDWB	45.7	46.1	53.2	58.3	67.4	74.4	78.7	76.5	69.7	60.1	53.2	49.6	(28)
(29)		2%	WB	42.0	41.8	44.0	49.5	57.0	64.4	69.9	69.1	63.7	56.2	49.2	45.1	(29)
(30)			MCDWB	43.6	43.7	48.2	54.6	62.6	71.1	75.6	73.4	67.6	58.0	50.6	46.5	(30)
(31)		5%	WB	40.0	39.9	41.8	47.2	55.0	62.3	68.2	67.7	61.9	54.6	47.6	43.2	(31)
(32)			MCDWB	41.2	41.8	45.1	51.5	60.0	68.1	73.1	71.9	65.2	56.4	48.8	44.6	(32)
(33)		10%	WB	38.4	38.2	40.1	45.0	53.2	60.5	66.6	66.2	60.2	53.1	46.3	41.7	(33)
(34)			MCDWB	39.7	40.1	42.9	48.7	57.8	65.7	71.3	70.3	63.4	54.9	47.5	43.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.0	5.7	7.4	8.5	9.8	11.1	10.5	9.9	8.1	6.0	4.8	4.9	(35)
(36)			MCDBR	6.4	7.8	12.1	14.2	14.8	16.0	14.6	12.7	11.5	8.1	6.1	6.3	(36)
(37)			MCWBR	5.9	6.2	8.0	8.7	8.1	8.5	7.1	6.5	6.6	5.8	5.5	5.9	(37)
(38)		5% WB	MCDWB	6.3	7.7	11.5	13.3	13.3	14.2	12.3	10.9	10.7	7.5	5.9	5.9	(38)
(39)			MCWBR	6.0	6.5	8.0	8.8	8.2	8.5	6.9	6.3	6.8	5.8	5.5	5.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.299	0.330	0.345	0.382	0.378	0.369	0.398	0.390	0.359	0.349	0.329	0.291	(40)	
(41)		taud	2.288	2.325	2.361	2.318	2.407	2.469	2.406	2.419	2.483	2.412	2.293	2.227	(41)	
(42)		Ebn at Noon	180	226	256	263	272	277	265	258	251	217	164	145	(42)	
(43)		Edh at Noon	16	23	29	35	34	32	34	31	25	21	16	12	(43)	
(44)	All-Sky Solar Radiation	RadAvg	115	297	780	1343	1783	1946	1767	1423	963	429	157	84	(44)	
(45)		RadStd	19	63	72	150	138	95	169	122	109	69	29	12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.68	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (38 neighbors)	+0.70	N/A	+2.47	N/A	N/A	N/A	-218	-283	N/A	N/A

Nomenclature: See separate page